DANIEL SANTANDER URRUTIA

Environmental Geophysicist & Catchment Modeler • Dual European M.Sc. Candidate

FULL TECHNICAL & POLICY CV (PAGE 1 OF 2)

📍 **Kiel, Germany** & **Santiago, Chile** ✉ **danisantander.urrutia@gmail.com** 🌐 **danisantander.github.io/Portafolio**

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Comprehensive Profile: Quantitative environmental scientist and geophysicist specializing in catchment water balance, subsurface electrical resistivity tomography (ERT), and greenhouse gas flux dynamics in rewetted peatlands. Holds a dual European Master of Science degree (CAU Kiel, Germany and AMU Poznań, Poland) alongside a 6-year engineering geophysics degree with Highest Distinction from Universidad de Chile. Combines rigorous experimental field campaigns with computational modeling (Python, R, QGIS, PostGIS, Docker), legislative policy advisory before Chile's National Congress, and large-scale public outreach (+700k reach).

OFFICIAL HIGHER EDUCATION & UNIVERSITY DEGREES

3 ACCREDITED DEGREES

M.Sc. in Environmental Management — **Christian-Albrechts-Universität zu Kiel (CAU Kiel)** — Kiel, Germany | 2023 – 2026

Faculty of Agricultural & Nutritional Sciences. Specialization: Catchment hydrology, soil hydro-physics, peatland rewetting mechanics, and atmospheric micrometeorology.

- **Master Thesis:** Biophysical modeling and multi-sensor monitoring of the Wallener Au rewetted peatland. Calibrated soil precompression stress (σ_p = 62.4 kPa), hydraulic conductivity (k_f), and water retention curves (pF) against water table fluctuations.
- **Digital Twin Innovation:** Built the *Wetland Monitor* 3D spatial web application, coupling real-time piezometer sensor streams with Sentinel-2 spectral indices (NDVI, NDWI) and Eddy Covariance carbon flux models (CO_2, CH_4).

M.Sc. in Environmental Protection — **Adam Mickiewicz University (AMU Poznań)** — Poznań, Poland | 2023 – 2026

Faculty of Geographical & Geological Sciences. Specialization: Freshwater ecology, limnology, aquatic ecosystem restoration, and EU environmental directives.

- **Research Focus:** Carbon sequestration dynamics in post-glacial lake basins and degraded fen peatlands. Microscopic taxonomy of freshwater bioindicators (diatoms, macrophytes) for water quality classification under the EU Water Framework Directive (WFD).

Geophysicist & B.Sc. in Geophysics (6-Year Engineering Degree) — **Universidad de Chile** — Santiago, Chile | 2015 – 2021

Faculty of Physical & Mathematical Sciences (FCFM). **Awarded with Highest Distinction (Distinción Máxima)**.

- Rigorous curriculum in solid Earth physics, continuum mechanics, potential field methods (gravity/magnetics), seismology, and applied electromagnetic exploration.
- **Engineering Thesis:** 2D/3D numerical modeling of unconfined aquifer depletion in the Maule River Basin under intensive agro-industrial forestry and mega-drought, utilizing Vertical Electrical Sounding (VES/SEV) and regional piezometric records.

SPECIALIZED POSTGRADUATE FELLOWSHIPS, SUMMITS & DIPLOMAS

INTERNATIONAL COHORTS

WSL SWISS FEDERAL RESEARCH INSTITUTE (2025)
BlueGreen Biodiversity Summit — Davos, Switzerland: Alpine wetland hydrology, riparian biodiversity indicators, and catchment climate resilience modeling.

IOER LEIBNIZ INSTITUTE / DRESDEN (2024–2025)
Interlinked Justice & Resilient Landscapes: Spatial planning, territorial equity, polycentric water governance, and post-carbon ecological transitions.

DEAL DOUGHNUT ECONOMICS ACTION LAB (2025)
Practitioner Training — Copenhagen, Denmark: Planetary boundaries downscaling, social foundations, and regenerative economic modeling for municipalities.

CR2 CENTER FOR CLIMATE & RESILIENCE (2023)
3rd Summer School: Resilient Coasts (57 hrs) — Chile: Directed by Dr. René Garreaud. Coastal climate hazard modeling, sea level rise, and adaptive shoreline planning.

SELECTED APPLIED RESEARCH & TECHNICAL PROJECTS

EMPIRICAL SCIENCE

Wallener Au Peatland Rewetting & Carbon Flux Dynamics — Schleswig-Holstein, Germany | 2023 – Present

Integrated continuous Eddy Covariance tower telemetry (LI-COR LI-7500/7700) with dynamic soil closed chamber campaigns measuring net ecosystem exchange (NEE), ecosystem respiration (R_{eco}), and biogenic methane fluxes. Analyzed soil precompression stress under heavy agricultural machinery vs. rewetted conditions.

AgriTwin IoT & Satellite Vineyard Water Telemetry Platform — Maule Basin, Chile | 2022 – 2023

Coupled LoRaWAN in-situ soil moisture and matric potential sensors with automated Sentinel-2 multispectral vegetation vigor pipelines (NDVI, NDRE) for precision deficit irrigation scheduling across Mediterranean viticultural catchments.

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DANIEL SANTANDER URRUTIA

Policy Advisory, Publications, Field Instrumentation & Strategic Communications

FULL TECHNICAL & POLICY CV (PAGE 2 OF 2)

PUBLIC POLICY ADVISORY, LEGISLATIVE TESTIMONY & GOVERNANCE

IMPACT & ADVOCACY

Parliamentary Expert Testimony — Chamber of Deputies of Chile (National Congress) — Valparaíso, Chile | 2023

Invited technical expert witness before the *Special Commission on Emergencies, Disasters and Firefighters*. Presented quantitative evidence linking industrial pine and eucalyptus monocultures to accelerated wildfire velocity, fuel connectivity, and severe groundwater depletion in rural catchments.

Parliamentary Bilateral Policy Column — German Green MP Dirk Kock-Rohwer & Comestible.info — Kiel / Schleswig-Holstein | 2024

Co-authored political ecology policy brief with Schleswig-Holstein Green MP Dirk Kock-Rohwer advocating for direct public subsidies for organic soil regeneration, non-chemical farming transitions, and legal protections for humic peat soils.

International Biomass & Carbon Accounting Investigation — Environmental Paper Network (EPN) — International | 2023 – 2024

Authored international investigative monograph *"Spotlight on Burning Biomass"* (Parral, Chile), exposing regulatory loopholes under the European Union Renewable Energy Directive (RED III) and corporate greenwashing in forest biomass burning.

SELECTED PUBLICATIONS, INVESTIGATIVE REPORTS & ESSAYS

POLICY & PRESS

- **Environmental Paper Network & Global Forest Coalition (2023):** *"Spotlight on Burning Biomass for Energy in South America: Community Opposition to a Power Plant in Parral, Chile."* Lead Technical Author.
- **Tomate Rojo & Plurinational Wetland Network (2023):** *"Waters Under Fire: What Do We Know and What Is Chile Doing?"* (D. Santander, L. Muñoz, C. Bonifetti). In-depth investigative report on wildfire impacts on aquatic and peatland systems.
- **El Desconcierto (2025):** *"Countries Too Rich to Grow? Norway and the World Degrowth Summit."* International dispatch from Oslo analyzing post-growth economic models, resource ceilings, and Nordic ecological transitions.
- **Observatorio Latinoamericano de Conflictos Ambientales - OLCA (2025):** *"Being Free on the Brink of the Abyss."* Four-part philosophical-political treatise on ecological collapse, energy boundaries, and territorial autonomy.
- **El Regionalista & Fundación Suelo (2023):** *"How Do We Prepare for a Climate Collapse?"* (C. Miranda & D. Santander). Policy brief on deep community adaptation, decentralized food systems, and soil biology.

STRATEGIC COMMUNICATIONS, DIGITAL ARCHITECTURE & MASS CAMPAIGNS

+700K REACH

Apruebo Rural & Primera Línea Prensa — Executive Communications Lead — Chile | 2021 – 2023

Conceived and coordinated a nationwide transmedia communication strategy with organic viral reach exceeding 700,000 citizens. Directed content production, live stage events, visual infographics, and decentralized rural assemblies.

Cuarentena Rap & Rap Beauchef — Founder, Artistic Director & Community Architect — Santiago & Global | 2020 – 2022

Engineered a global digital spoken-word and hip-hop league via Discord and streaming platforms during pandemic lockdowns, managing an active international community of +12,000 participants across 8 countries.

COMPLETE TECHNICAL STACK & FIELD INSTRUMENTATION MATRIX

LABORATORY & SOFTWARE

PROGRAMMING & GIS

- Python (GeoPandas, Rasterio, NumPy, SciPy)
- R (sf, ggplot2, spatstat, EDA)
- QGIS & ArcGIS Pro (LiDAR, DEMs)
- Google Earth Engine (GEE)
- PostgreSQL / PostGIS • Docker • Git

FIELD EQUIPMENT

- Syscal Pro (Multi-electrode ERT)
- LI-COR LI-7500/7700 (Eddy Covariance)
- Dynamic soil closed flux chambers
- CTD divers & piezometers (water level)
- Edelman undisturbed soil coring

FULL-STACK & WEB

- TypeScript, React, Next.js
- FastAPI & RESTful APIs
- Three.js & Unity 3D spatial twins
- Delft3D hydrodynamic flood models
- LaTeX / Typst mathematical typesetting

Working Languages: Spanish (Native — parliamentary testimony & editorial prose) • **English** (C1 Fluent — international keynote speaking & scientific publishing) • **German** (B1-B2 Working — CAU Kiel graduate coursework & technical dialogue).

TAILORED DOSSIER NOTICE This 2-page curriculum reflects a general comprehensive baseline. Specific project portfolios, full code repositories, and customized dossiers are assembled upon request.

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